Cisco PE Node MALL-PE-08 Hardware/Environmental Fault SOP on Malleswaram Node

Node: MALL-PE-08 | **Vendor:** Cisco | **Severity:** Major | **Alarm Code:** ALARM_2837

Description

The Cisco PE Node MALL-PE-08 hardware/environmental fault SOP is designed to identify and resolve issues related to ASIC memory parity errors, line card crashes, and power supply redundancy loss on the Malleswaram Node, which is a critical component of the Telecom network infrastructure. This SOP is based on the Cisco IOS CLI dialect and is intended for Senior MTS personnel with extensive experience in Cisco PE node management. The underlying protocol mechanics and hardware components involved in this fault domain include the Cisco IOS software operating on the ASR 9000 series routers, which utilize the line card ASICs to process network traffic. The power supply redundancy loss is a critical failure scenario, as it can lead to a complete loss of network connectivity to the node. The ASIC memory parity errors and line card crashes can be caused by a variety of factors, including hardware degradation, software bugs, or environmental stressors such as high temperatures or power surges. To resolve these issues, this SOP outlines a structured approach to troubleshooting and repair, which includes physical checks, CLI-based diagnostics, and configuration changes.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Malleswaram Node to perform physical checks and repairs.

- Verify the status of the redundant links to ensure that they are operational and not contributing to the fault.

- Review the Cisco IOS software version and configuration to ensure that they are up-to-date and consistent with best practices.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Errors	High
Line Card Crashes	Medium
Power Supply Redundancy Loss	High

Diagnostic Commands & Expected Outputs

```
Command: show tech-support, Expected Output: Technical support information for the Cisco IOS software and hardware components., Description: This command gathers detailed information about the Cisco IOS software and hardware components, including the ASIC memory parity errors, line card crashes, and power supply redundancy loss.
```

```
Command: show platform, Expected Output: Platform information, including the line card and power supply status., Description: This command provides a detailed view of the platform, including the status of the line card and power supply, which can help identify the root cause of the fault.

Command: show environment, Expected Output: Environmental information, including temperature and humidity levels., Description: This command provides information about the environmental conditions at the Malleswaram Node, which can help identify potential stressors that may be contributing to the fault.
```

Resolution Steps

Step 1: Step: Physical Checks and Repairs, Command: physical-checks-and-repairs, Description: Perform physical checks on the Malleswaram Node to identify any issues with the hardware or environment that may be contributing to the fault. Perform repairs as necessary to ensure that the node is operational and not contributing to the fault.

Step 2: Step: Software Upgrades and Configuration Changes, Command: software-upgrades-and-configuration-changes, Description: Review the Cisco IOS software version and configuration to ensure that they are up-to-date and consistent with best practices. Upgrade the software or make configuration changes as necessary to resolve the fault.

Step 3: Step: Hardware Replacement, Command: hardware-replacement, Description: Replace the faulty hardware components, including the line card and power supply, with new or refurbished parts to ensure that the node is operational and not contributing to the fault.

Post-Check Verification

```
Command: show tech-support, Expected Output: Technical support information for the Cisco IOS software and hardware components., Description: Verify that the technical support information indicates that the fault has been resolved, and that the Cisco IOS software and hardware components are functioning correctly.

Command: show platform, Expected Output: Platform information, including the line card and power supply status., Description: Verify that the platform information indicates that the line card and power supply are functioning correctly, and that the fault has been resolved.
```

Roll-back Procedures

- Step: Revert Software Changes, Command: revert-software-changes, Description: Revert any software changes made during the resolution process to ensure that the node returns to its original configuration.

- Step: Reinstall Faulty Hardware, Command: reinstall-faulty-hardware, Description: Reinstall the faulty hardware components, including the line card and power supply, to ensure that the node returns to its original configuration.